```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

df = pd.read_csv("/content/air_quality_dataset -
air_quality_dataset.csv")

conditions = [
    (df["PM2.5"] <= 35),
    (df["PM2.5"] > 35) & (df["PM2.5"] <= 75),
    (df["PM2.5"] > 75)
]

labels = ["Good", "Moderate", "Poor"]


df["air_quality_dataset"] = np.select(conditions, labels,
default="Unknown")
X = df.drop(columns=["air_quality_dataset"])
y = df["air_quality_dataset"]

print("Feature Names:", list(X.columns))
print("Target Classes:", y.unique())
print("Number of Samples:", X.shape[0])
print("Number of Features:", X.shape[1])
print(df.head())
print("\nClass Distribution")
print(y.value_counts())

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=0,
    stratify=y
)
print("Shape of X_train:", X_train.shape)
print("Shape of X_test:", X_test.shape)
model = LogisticRegression(max_iter=500)
model.fit(X_train, y_train)
print("Model Trained Successfully!")
y_pred = model.predict(X_test)
results = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})
print(results.head(10))
accuracy = accuracy_score(y_test, y_pred)
```

```
print(f"\nTest Accuracy: {accuracy:.2%}")


Feature Names: ['PM2.5', 'PM10', 'NOx', 'NO2', 'SO2', 'VOCs', 'CO',
'CO2', 'CH4', 'Temperature']
Target Classes: ['Moderate' 'Poor' 'Good']
Number of Samples: 500
Number of Features: 10
        PM2.5        PM10         NOx        NO2        SO2
VOCs  \
0   39.967142   57.926035  116.192213  55.230299   4.531693
75.317261
1  101.935672  150.774299   76.826826  79.051618  18.744780
145.083987
2   70.996192  138.948796  158.731020  60.466604  14.892239
145.147338
3   28.464728   63.643900   25.385343  15.333286   7.647429
130.022319
4   78.265276  113.977926  105.644340  59.202337  17.696806
181.713667

         CO         CO2       CH4  Temperature air_quality_dataset
0  2.789606  427.674347  1.706105    31.085120            Moderate
1  1.966569  529.739619  2.492663    33.711103                Poor
2  2.626446  499.889443  2.431165    33.778698            Moderate
3  1.779360  388.283712  1.818563    31.565877                Good
4  3.240533  464.739197  2.597225    32.229835                Poor

Class Distribution
air_quality_dataset
Moderate    285
Poor        115
Good        100
Name: count, dtype: int64
Shape of X_train: (400, 10)
Shape of X_test: (100, 10)
Model Trained Successfully!
     Actual Predicted
0      Good      Good
1      Poor      Poor
2      Poor      Poor
3  Moderate  Moderate
4      Good      Good
5      Good      Good
6      Poor      Poor
7  Moderate  Moderate
8  Moderate  Moderate
9      Poor      Poor

Test Accuracy: 94.00%
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/linear_model/
_logistic.py:465: ConvergenceWarning: lbfgs failed to converge
(status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as
shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-
regression
  n_iter_i = _check_optimize_result(
```

```
new_sample = [[40, 60, 90, 45, 12, 1.8, 0.8, 420, 2.1, 30]]
prediction = model.predict(new_sample)
print("Predicted Air Quality:", prediction[0])
```

```
Predicted Air Quality: Moderate
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/utils/
validation.py:2739: UserWarning: X does not have valid feature names,
but LogisticRegression was fitted with feature names
  warnings.warn(
```